

Polynomial Inequalities

Solve each inequality.

1) $(x - 4)(x + 3) < 0$

2) $(x - 4)(x + 1) \geq 0$

3) $(x - 1)(3x - 4) \geq 0$

4) $(x + 8)(x + 2)(x - 3) \geq 0$

5) $x^2 + 5x + 4 \leq 0$

6) $x^2 - 14x + 49 \geq 0$

7) $x^2 - 4x - 32 > 0$

8) $x^2 + 16x + 24 > 6x$

9) $(x + 5)(x - 2)(x - 1)(x + 1) < 0$

10) $(x + 8)^2(x + 5)(x + 7)^2 \geq 0$

Critical thinking question:

11) Write a polynomial inequality with the solution: $\{-1\} \cup \{2\} \cup [3, \infty)$